CampusWatch

The campus topology in Cisco Packet Tracer

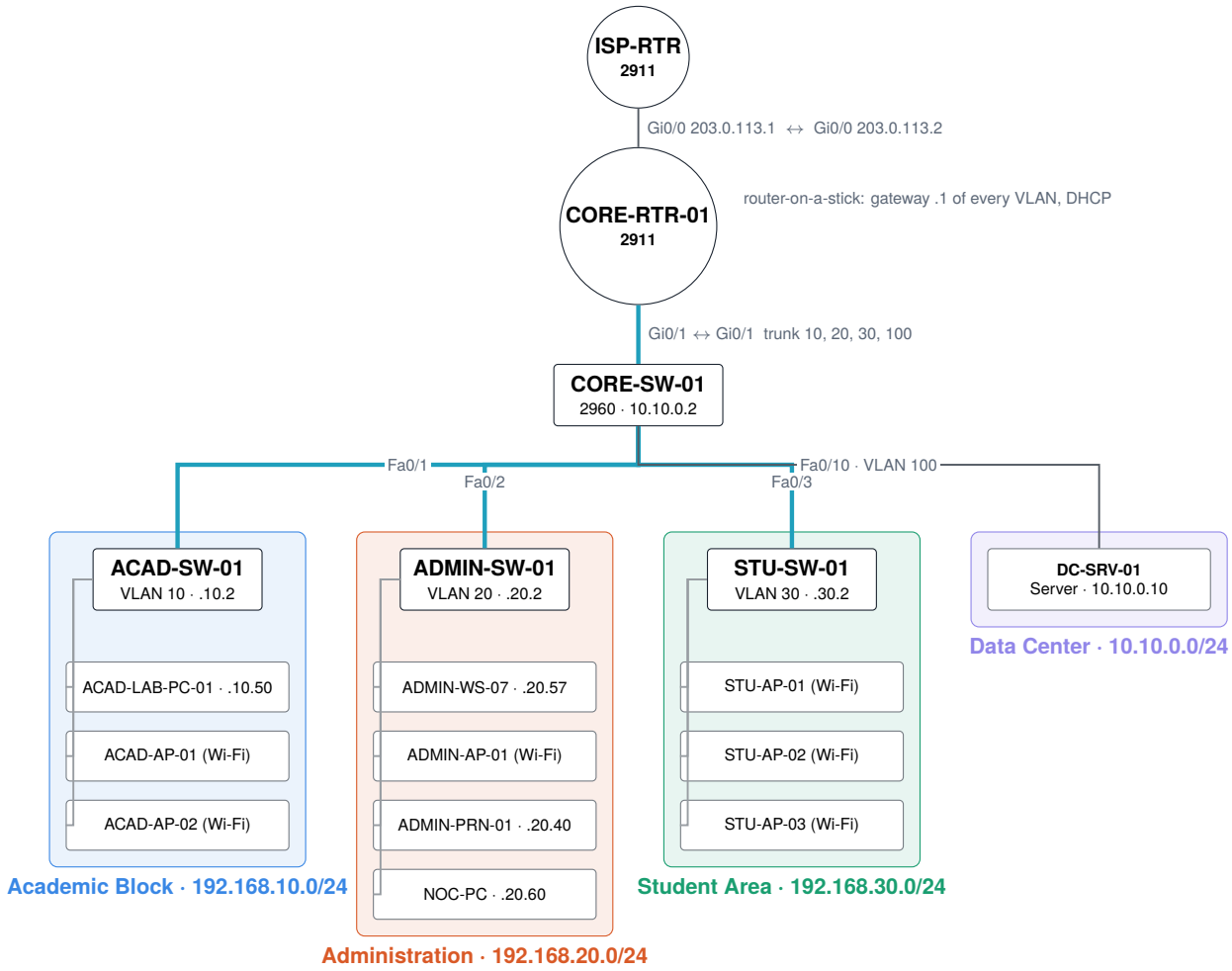
Devices, cabling, IOS configurations and verification for Packet Tracer 8.x

This guide builds the CampusWatch campus in Packet Tracer. It uses the dashboard's demo device names, VLANs and IP addresses, so you can show the same network in both. Allow 30 to 40 minutes. Paste each configuration block into the device's **CLI** tab (or **Config** → **CLI**).

Why the dashboard does not poll Packet Tracer

Packet Tracer is a simulation: its devices exist only inside the program, so CampusWatch running on your laptop cannot reach them. Demonstrate SNMP from inside Packet Tracer with a PC's **MIB Browser** (Section 6), then show the dashboard in Demo mode, which uses the same campus and labels its data as synthetic.

1 Topology



Campus area	VLAN	Subnet	Gateway	Switch mgmt	DHCP range / colour
Academic Block	10	192.168.10.0/24	.1	.2	.100 to .254, blue
Administration Block	20	192.168.20.0/24	.1	.2	.100 to .254, orange
Student Area	30	192.168.30.0/24	.1	.2	.100 to .254, green
Data Center	100	10.10.0.0/24	.1	.2	static only, violet

Static addresses (outside the DHCP range): switches `.2`, lab PC `.50`, workstation `.57`, printer `.40`, NOC PC `.60`, server `10.10.0.10`. The colours match the dashboard's colour for each area.

2 Place devices and cables

Device (rename to this)	Packet Tracer model
CORE-RTR-01, ISP-RTR	Routers → **2911**
CORE-SW-01, ACAD-SW-01, ADMIN-SW-01, STU-SW-01	Switches → **2960-24TT**
DC-SRV-01	End Devices → **Server-PT**
ACAD-LAB-PC-01, ADMIN-WS-07, NOC-PC	End Devices → **PC-PT**
ADMIN-PRN-01	End Devices → **Printer-PT**
ACAD-AP-01/02, ADMIN-AP-01, STU-AP-01/02/03	Network Devices → Wireless → **AccessPoint-PT**
Wireless clients (2 or 3 per area)	**Laptop-PT** with its NIC swapped for **WPC300N** (power off, drag the module out, drag WPC300N in, power on), or **Smartphone-PT**

From	Port	To	Port	Cable
ISP-RTR	Gi0/0	CORE-RTR-01	Gi0/0	cross-over
CORE-RTR-01	Gi0/1	CORE-SW-01	Gi0/1	straight-through
CORE-SW-01	Fa0/1	ACAD-SW-01	Gi0/1	cross-over
CORE-SW-01	Fa0/2	ADMIN-SW-01	Gi0/1	cross-over
CORE-SW-01	Fa0/3	STU-SW-01	Gi0/1	cross-over
CORE-SW-01	Fa0/10	DC-SRV-01	Fa0	straight-through
ACAD-SW-01	Fa0/2–4	lab PC, ACAD-AP-01, ACAD-AP-02	Fa0 / Port 0	straight-through
ADMIN-SW-01	Fa0/2–5	ADMIN-WS-07, ADMIN-AP-01, ADMIN-PRN-01, NOC-PC	Fa0 / Port 0	straight-through
STU-SW-01	Fa0/2–4	STU-AP-01, STU-AP-02, STU-AP-03	Port 0	straight-through

Presentation polish

Use **Place Note** to label each area with its VLAN and subnet, and **Draw Rectangle** around each area in the colour shown above, so the Packet Tracer view and the dashboard look like the same campus.

3 Router configurations

CORE-RTR-01

```
enable
configure terminal
hostname CORE-RTR-01
no ip domain-lookup
!
interface GigabitEthernet0/0
 description ISP uplink
 ip address 203.0.113.2 255.255.255.252
 no shutdown
!
interface GigabitEthernet0/1
 description Trunk to CORE-SW-01
```

```
 no ip address
 no shutdown
!
interface GigabitEthernet0/1.10
 description Academic Block
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
!
interface GigabitEthernet0/1.20
 description Administration Block
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
!
interface GigabitEthernet0/1.30
 description Student Area
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
!
interface GigabitEthernet0/1.100
 description Data Center
 encapsulation dot1Q 100
 ip address 10.10.0.1 255.255.255.0
!
ip dhcp excluded-address 192.168.10.1 192.168.10.99
ip dhcp excluded-address 192.168.20.1 192.168.20.99
ip dhcp excluded-address 192.168.30.1 192.168.30.99
!
ip dhcp pool ACADEMIC
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 10.10.0.10
ip dhcp pool ADMIN
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 10.10.0.10
ip dhcp pool STUDENT
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 10.10.0.10
!
ip route 0.0.0.0 0.0.0.0 203.0.113.1
!
snmp-server community campuswatch-ro RO
end
write memory
```

ISP-RTR (optional, simulates the Internet)

```
enable
configure terminal
hostname ISP-RTR
no ip domain-lookup
interface GigabitEthernet0/0
 ip address 203.0.113.1 255.255.255.252
 no shutdown
interface Loopback0
 description Simulated Internet host
```

```
 ip address 8.8.8.8 255.255.255.255
ip route 192.168.0.0 255.255.0.0 203.0.113.2
ip route 10.10.0.0 255.255.255.0 203.0.113.2
end
write memory
```

4 Switch configurations

CORE-SW-01

```
enable
configure terminal
hostname CORE-SW-01
no ip domain-lookup
vlan 10
 name ACADEMIC
vlan 20
 name ADMIN
vlan 30
 name STUDENT
vlan 100
 name DATACENTER
exit
interface GigabitEthernet0/1
 description Trunk to CORE-RTR-01
 switchport mode trunk
interface range FastEthernet0/1 - 3
 description Trunks to building switches
 switchport mode trunk
interface FastEthernet0/10
 description DC-SRV-01
 switchport mode access
 switchport access vlan 100
 spanning-tree portfast
interface Vlan100
 ip address 10.10.0.2 255.255.255.0
 no shutdown
exit
ip default-gateway 10.10.0.1
snmp-server community campuswatch-ro RO
end
write memory
```

ACAD-SW-01

```
enable
configure terminal
hostname ACAD-SW-01
no ip domain-lookup
vlan 10
 name ACADEMIC
exit
interface GigabitEthernet0/1
 description Uplink to CORE-SW-01
```

```
 switchport mode trunk
interface range FastEthernet0/1 - 24
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
interface Vlan10
 ip address 192.168.10.2 255.255.255.0
 no shutdown
exit
ip default-gateway 192.168.10.1
snmp-server community campuswatch-ro RO
end
write memory
```

ADMIN-SW-01 and STU-SW-01

Copy the ACAD-SW-01 block and replace these values (`vlan`, `switchport access vlan`, `interface VlanXX`, the IP address and the default gateway):

Switch	hostname	VLAN id / name	Vlan interface IP	Default gateway
ADMIN-SW-01	`ADMIN-SW-01`	`20 / ADMIN`	`192.168.20.2`	`192.168.20.1`
STU-SW-01	`STU-SW-01`	`30 / STUDENT`	`192.168.30.2`	`192.168.30.1`

5 End devices

Device	Where	Settings
DC-SRV-01	Desktop → IP Configuration	`10.10.0.10` / `255.255.255.0`, gateway `10.10.0.1`, DNS `10.10.0.10`
DC-SRV-01	Services → HTTP	On (optionally edit `index.html` to say "Campus LMS")
DC-SRV-01	Services → DNS	On; add A record `lms.campus.local` → `10.10.0.10`
ACAD-LAB-PC-01	Desktop → IP Configuration	`192.168.10.50`, gateway `.1`, DNS `10.10.0.10`
ADMIN-WS-07	Desktop → IP Configuration	`192.168.20.57`, gateway `.1`, DNS `10.10.0.10`
ADMIN-PRN-01	Config → FastEthernet0, Settings	`192.168.20.40` / `255.255.255.0`; gateway `192.168.20.1`
NOC-PC	Desktop → IP Configuration	`192.168.20.60`, gateway `.1`, DNS `10.10.0.10`
ACAD-AP-01/02	Config → Port 1	SSID `CAMPUS-ACAD`, WPA2-PSK, passphrase `academic123`
ADMIN-AP-01	Config → Port 1	SSID `CAMPUS-ADMIN`, WPA2-PSK, passphrase `admin1234`
STU-AP-01/02/03	Config → Port 1	SSID `CAMPUS-STUDENT`, WPA2-PSK, passphrase `student123`
Laptops / phones	Config → Wireless0, then IP Configuration	the area's SSID and passphrase; IP set to **DHCP**

Note

AccessPoint-PT is a layer-2 bridge with no management IP in Packet Tracer, so the access point addresses shown in the dashboard (`.11`, `.12`, `.13`) are inventory labels only. The Wi-Fi passphrases are lab placeholders; change them.

6 Verify (show these in the presentation)

On **CORE-RTR-01** and the switches:

```
show ip interface brief        ! all sub-interfaces up/up
show vlan brief                ! access ports in the right VLAN
show interfaces trunk          ! trunks carry 10,20,30,100
show ip dhcp binding           ! laptops and phones received addresses
```

From end devices (Desktop → Command Prompt):

```
ping 192.168.10.1              ! own gateway
ping 10.10.0.10                ! inter-VLAN routing to the server
ping 192.168.30.100            ! another area (first DHCP lease in Student Area)
ping 8.8.8.8                   ! "Internet" (if ISP-RTR is built)
```

Web browser on any client: `http://lms.campus.local` (tests DNS and HTTP on DC-SRV-01).

SNMP inside Packet Tracer

On **NOC-PC**: Desktop → **MIB Browser** → Advanced. Address `192.168.10.2` (ACAD-SW-01), read community `campuswatch-ro`, version v2c, OK. Expand the MIB tree to `system` → `sysName` or `sysUpTime` and press **GET**. Repeat for `10.10.0.2` or `192.168.20.1`. How much of the MIB tree is available depends on the Packet Tracer version, which is why CampusWatch's SNMP mode targets real devices or an SNMP simulator instead.

Traffic for the story

Switch Packet Tracer to **Simulation** mode, filter on ICMP and HTTP, then send pings or browser requests to show packets crossing the trunk and being routed between VLANs.

7 Map to the dashboard

Packet Tracer	CampusWatch demo device	Address
CORE-RTR-01	CORE-RTR-01 (Data Center)	10.10.0.1, gateway of every VLAN
DC-SRV-01	DC-SRV-01 "LMS server"	10.10.0.10
ACAD-SW-01, ACAD-AP-01/02, ACAD-LAB-PC-01	same names (Academic Block)	192.168.10.2 / .11 / .12 / .50
ADMIN-SW-01, ADMIN-AP-01, ADMIN-WS-07, ADMIN-PRN-01	same names (Administration Block)	192.168.20.2 / .11 / .57 / .40
STU-SW-01, STU-AP-01/02/03	same names (Student Area)	192.168.30.2 / .11 / .12 / .13
CORE-SW-01, ISP-RTR, NOC-PC	not in the demo inventory	10.10.0.2, 203.0.113.1, 192.168.20.60
(not built)	DC-FW-01 firewall	optional: an ASA 5506-X between CORE-RTR-01 Gi0/0 and ISP-RTR

Suggested flow: show the Packet Tracer topology, run the pings and the MIB Browser GET, then open CampusWatch. Its front panel shows the same switches, access points and VLANs, with synthetic traffic, and says so on screen.

Save the Packet Tracer file as `campus-topology.pkt` (File → Save As) in the project's `docs/` folder so it ships with the project.